

Sourav Kumar

+91 7463959457 | 2204rajsourav@gmail.com | [LinkedIn](#) | [Portfolio](#)

Profile Summary

- Machine Learning Engineer with hands-on experience building production-grade AI systems and LLM-powered applications using RAG pipelines.
- Built Enterprise-FraudShield-AI processing 10,000+ transactions/sec at 94% accuracy; proficient in Python, TensorFlow, Kafka, LangChain, data pipelines, and AWS.
- IBM-certified ML professional with Microsoft Azure AI-900 and CodeWithHarry Python credentials; available from May 2026.

Technical Skills

Languages: Python, Java, SQL, C, C++

AI / ML / NLP: LangChain, RAG, LLMs, Prompt Engineering

ML Frameworks: Scikit-learn, XGBoost, TensorFlow, LSTM, ARIMA, Prophet, Random Forest

Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, Apache Kafka, Redis, Git

Backend & Evaluation: FastAPI, REST APIs, Hyperparameter Tuning, Feature Engineering, ROC-AUC, RMSE

Experience

IBM June 2025 – August 2025

Machine Learning Intern [View Certificate](#)

Remote

- Preprocessed 10,000+ healthcare records, reducing inconsistencies by 30%; built end-to-end predictive pipeline achieving 87% accuracy on insurance charge prediction.
- Reduced prediction error by 15% via hyperparameter tuning and cross-validation across Linear Regression, Decision Tree, and Random Forest models.
- Visualised key insights and model performance using Matplotlib and Seaborn, presenting findings to the project team.

Samadhaan Foundation

June 2023 – August 2023

Data Analysis Volunteer [View Certificate](#)

On-site

- Analyzed data from 200+ beneficiaries using Python and Excel; built dashboards improving reporting efficiency by 25% and automated manual data entry processes.
- Conducted data analysis and supported digital management initiatives to improve organisational efficiency.

Projects

Enterprise-FraudShield-AI | *Python, FastAPI, Kafka, XGBoost, Redis, Docker* | [Github](#) Dec 2025

- Engineered real-time fraud detection processing 10,000+ transactions/sec with 94% accuracy, reducing latency by 60%; estimated \$2-5M annual fraud loss prevention.
- Implemented XGBoost, anomaly detection, and behavioral analytics to detect refund abuse; deployed with Docker, Uvicorn, and async REST APIs.

Stock Price Prediction Models | *Python, LSTM, ARIMA, Prophet, TensorFlow* | [Github](#) Aug – Sep 2025

- Benchmarked LSTM, ARIMA, and Prophet across 5 stock datasets; ensemble methods achieved 22% accuracy improvement; built EDA dashboards visualizing trends, volatility, and model performance metrics.
- Performed hyperparameter tuning and cross-validation to reduce overfitting, achieving stable forecasting outcomes across 5 different stock datasets.

Fish Species Classifier (Transfer Learning) | *Python, MobileNetV3, TensorFlow, OpenCV* | [Github](#) May 2024

- Trained MobileNetV3 on 9,000 images across 9 classes; achieved 90% accuracy, reduced classification time by 85%; used 80/20 split, data augmentation, and confusion matrix evaluation.
- Built data augmentation and preprocessing pipelines (80/20 train-validation split) ensuring robust generalisation across varied datasets.

Education

University of Petroleum and Energy Studies (UPES), Dehradun

Aug 2022 – May 2026

B.Tech, Computer Science & Engineering — AI/ML Specialization

Extracurricular / Certifications

- IBM Certified:** Healthcare Cost Prediction using Machine Learning
- Microsoft Certified:** Azure AI Fundamentals (AI-900)
- Complete 2025 Python Bootcamp** — CodeWithHarry